Free-history timescape: a data-driven void history reconciles the supernova–BAO–CMB geometry but does not resolve the Hubble tension

Viacheslav Zhygulin[1,*]

[1] *Independent Researcher*

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Paper I of this program showed that the timescape cosmology in its standard one-parameter tracker closure cannot fit type Ia supernovae (SNe), baryon acoustic oscillations (BAO), and the cosmic microwave background (CMB) acoustic scale together: the SNe drive the present void fraction toward $f_{v0} \simeq 0.85$ while BAO+CMB demand $\simeq 0.64$ (a 4.8–6.6σ split), yet a model-independent free-$E(z)$ check shows the data are not internally contradictory—the split is the tracker shape's rigidity, not a data conflict. Here we free the void-fraction history $f_v(z)$ from the attractor, keeping the Buchert two-phase average and the wall/void clock-rate dressing, and re-test end to end. *Part I (kinematic):* the freed $f_v(z)$ reconciles Pantheon+, DESI BAO, and the Planck acoustic scale at joint $\chi^2 = 1396.1$, below flat ΛCDM (1402.2) and far below the tracker (1469.3), at a physically plausible $f_{v0} = 0.640$; the amplitude split dissolves to 0.16σ (tracker: 6.5σ). This reconciliation uses the Pantheon+ stat+sys covariance; under the cosmology-independent covariance of Lane, Seifert et al. [17, 18] (the FLRW-fiducial bias correction removed) the supernova likelihood instead prefers ΛCDM to the fixed free-history at every redshift cut (ΔBIC = +11 to +27), so the kinematic shape-sufficiency is itself covariance-dependent. But this is the backreaction the Hubble diagram *wants*—a kinematic target that violates Buchert integrability, not a proven solution. *Part II:* four independent falsification tests each fail. The required history must decline by ×3.31 over $0 < z < 2.33$ while the observed BOSS DR12 void population declines only ×1.16 (the $z = 0$ level matches but the decline shape is unavailable, by a floor theorem); the predicted local expansion-rate bias is +2.4% (+1.6% lensing-physical) versus the +8.4% SH0ES requires—the full admissible band lies below the target (pre-registered PARTIAL; failing outright when the observed catalog is forced in); enforcing dynamical consistency collapses the three-phase Buchert solve to the tracker limit, missing the geometry model-comparison bar by 4953; and the in-model sound horizon is $r_d = 199.6$ Mpc (+36%), driving the global bare $\bar{H}_0$ to $\sim$32–40 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Removing the tracker's rigidity reconciles the distance geometry, but the flexibility is kinematic: free-history timescape does not resolve the Hubble tension. Consistent with Paper I, eliminating dark energy through inhomogeneity does not resolve the tension on present public data.

I. INTRODUCTION

The proposal that the apparent cosmic acceleration—and, by extension, the Hubble tension—reflects the inhomogeneity of the late Universe rather than a new energy component is the dark-energy-free inhomogeneity programme. The timescape cosmology of Wiltshire [6–8] is its most developed member: the present, void-dominated Universe is modelled as a two-phase Buchert spatial average [9, 10] of spatially-flat "walls" (bound structures, where observers sit) and negatively-curved "voids," with a quasilocal clock-rate gradient ("dressing") so that a wall observer fitting Friedmann–Lemaître–Robertson–Walker (FLRW) infers "dressed" parameters distinct from the volume-average "bare" ones. Apparent acceleration follows without dark energy. Recent Pantheon+ reanalyses with a cosmology-independent covariance report strong Bayesian evidence for timescape over ΛCDM [17, 18], reviving the question of whether the framework also addresses the Hubble tension.

This is the second paper of a three-paper arc. Paper I [1] subjected the standard timescape *tracker*—the one-parameter late-time attractor that fixes the void-fraction history $f_v(z)$—to a uniform SN+BAO+CMB harness and found that no single tracker fits the three geometry probes together: the supernovae pull the present void fraction toward $f_{v0} \simeq 0.85$ while the BAO+CMB geometry demands $f_{v0} \simeq 0.64$, a 4.8–6.6σ parameter split. Crucially, a model-independent free-$E(z)$ spline fit to the same three datasets matches ΛCDM: the data are *not* internally contradictory. Paper I therefore diagnosed the tracker's one-parameter rigidity—not the timescape mechanism, and not a genuine SN-versus-BAO data conflict—as the cause of the combined-fit failure.

The present paper asks the immediate follow-up question: *is the timescape mechanism right but its tracker closure wrong?* We free the void history $f_v(z)$ from the tracker attractor, retain the Buchert two-phase average and the wall/void clock-rate dressing unchanged, and re-test the framework end to end—first against the SN+BAO+CMB geometry, then against every deeper physical requirement the reconciling history must satisfy.

We state the two-part answer up front. **Part I** (Sec. III): freeing $f_v(z)$ *does* reconcile the geometry. The freed history fits Pantheon+, DESI DR1 BAO, and the Planck acoustic scale at joint $\chi^2 = 1396.1$, below flat ΛCDM (1402.2) and far below the tracker (1469.3), at a physically plausible present void fraction $f_{v0} = 0.640$; the

* s.zhygulin@gmail.com

SN-versus-BAO+CMB amplitude split dissolves from the tracker's 6.5σ to 0.16σ. Throughout we label this reconciliation *kinematic*: it is a well-defined target the Hubble diagram wants, but the forced history need not satisfy the Buchert integrability conditions and is not, on its face, a general-relativistic solution. **Part II** (Sec. IV): four independent falsification tests ask whether the reconciling history is physically *available*, dynamically *consistent*, and early-Universe *calibratable*. All four fail. The required history is not present in the observed void population (its decline is $\times 3.31$ against an observed $\times 1.16$; *shape-unavailable*); it predicts a local expansion-rate bias of $+2.4\%$ against the $+8.4\%$ required—the full admissible band below the target, a pre-registered PARTIAL that fails outright when the observed catalog is forced in (*fails*); enforced onto a dynamical three-phase Buchert solve it collapses to the two-phase tracker limit and misses the geometry bar by 4953 (*closed*); and its in-model sound horizon is $r_d = 199.6$ Mpc, $+36\%$ above the external ruler, driving the bare Hubble rate to $\sim$32–40 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (*fails*). This is *reconciliation without resolution*: removing the tracker's rigidity reconciles the distance geometry, but the flexibility is kinematic and closes at every physical test. The result corroborates and sharpens Paper I's conclusion.

Every quantitative claim below traces to a committed analysis script and its result file (Sec. VI); each verdict is stated against a pre-registered threshold quoted before the verdict.

II. FREE-HISTORY TIMESCAPE FRAMEWORK

We use only what is needed; the derivations are in [6, 7, 12] and the uniform harness in [1]. The void volume fraction $f_v(t)$ grows to a present value f_{v0} in a two-phase Buchert average of walls and voids [9, 10]. The lapse $\bar\gamma \equiv dt/d\tau_w$ relates volume-average time t to wall proper time τ_w; in the late-time tracker solution the dressed and bare Hubble parameters obey [7, eq. (B9)]

$$H_0 = g_{\rm dress}(f_{v0})\,\bar H_0, \qquad g_{\rm dress}(f_{v0}) = \frac{4f_{v0}^2 + f_{v0} + 4}{2(2+f_{v0})}, \tag{1}$$

with $\bar\gamma_0 = \frac{1}{2}(2+f_{v0})$. The tracker is the *one-parameter* closure that fixes the entire history $f_v(z)$ from f_{v0} alone, and has been fitted to multiple probes at a low dressed H_0 [11, 13]. We drop it: we let $f_v(z)$ be a free function on a redshift grid, solve the dressed geometry (luminosity and transverse-comoving distances, following the DNW13 machinery [12] and the dressed-distance implementation of [16]) for each admissible history, and later confront the freed history with the observed void population.

Freeing the history exposes a lapse-convention choice that the tracker hides. We carry two readings. *LA (algebraic)* evaluates the lapse from the tracker-form combination $\frac{1}{2}(2+f_v)$ node by node. *LB (rate-ratio)* builds the lapse from the local expansion-rate ratio, so it depends on the history's slope $f_v'(z)$ and fixes a self-consistent $\bar\gamma_0$ by shooting. LA is the primary reading; LB, together with the no-lapse control (V0), brackets the required history as a lapse-reading systematic used in Part II.

Solver validation gate. The general (non-tracker) dressed-geometry solver is a faithful superset of the validated tracker geometry: driven to the tracker history it reproduces the tracker limit to a maximum fractional transverse-distance error $\lesssim 6\times10^{-8}$ and an SN $\chi^2 = 1391.545$ (matching the Paper I tracker value), the LB gate passes at joint tracker $\chi^2 = 1469.293$ against the reference 1469.293, and it exhibits the genuine non-FLRW signature $dD_M/dz \neq D_H \equiv c/H(z)$, which collapses to equality only in the FLRW limit. This is the load-bearing statement that the machinery is correct before we ask what it prefers.

III. PART I: GEOMETRIC RECONCILIATION

All Part I results are *kinematic*: they establish that the timescape mechanism, with its history freed, has enough shape freedom to fit the distance geometry. Whether that shape is physically realised is the subject of Part II.

A. Data and harness

We reuse Paper I's uniform harness and pipeline validation [1]. The supernova data are the Pantheon+ release [19, 20]; the cosmology sample is the $N = 1580$ SNe with $z_{\rm HD} > 0.01$ that are not Cepheid calibrators ($0.010 < z < 2.26$), analysed under the full public statistical+systematic covariance with the absolute-magnitude/H_0 offset marginalised analytically (the supernova verdict has historically been sensitive to the light-curve fitter [14], which the uniform harness holds fixed across models). The main joint fit reuses Paper I's DR1 SN+BAO+CMB harness for direct comparability with Paper I; DESI DR2 [4, 5] is used as a cross-check (Sec. III C) and throughout Part II. The CMB enters only through the Planck 2018 acoustic-scale point [3]. As in Paper I, the sound-horizon ruler r_d enters BAO+CMB only through the distance amplitude, so the redshift *shape*—and hence the history $f_v(z)$—is independent of the r_d calibration (a fact we exploit, and whose limits we probe, in Sec. IV D).

B. The required void history (Probe R)

We fit the freed history jointly to SN+BAO+CMB and compare, on identical data, against the model-independent free-$E(z)$ spline, evolving dark energy (w_0w_aCDM, the DESI-hinted thawing direction [4]), flat ΛCDM, and the one-parameter tracker (Table I). The pre-registered thresholds [1] are: *reconciles* if joint $\chi^2 \leq$

TABLE I. Joint SN+BAO+CMB fit, identical data and harness: minimum χ^2 of each model. The freed history (headline row) fits below flat ΛCDM and far below the one-parameter tracker; only the model-independent free-$E(z)$ spline does better. The pre-registered reconcile threshold is joint $\chi^2 \leq 1412.24$. Column k is the fitted parameter count; column $\chi^2 - \chi^2_{\Lambda\mathrm{CDM}}$ is the difference from the ΛCDM row. Raw χ^2 is compared; no information-criterion preference is claimed. The free-history improvement over ΛCDM carries 4 extra parameters and is not significant (Wilks $p = 0.19$ on DR1, 0.82 on DR2; BIC favours ΛCDM, ΔBIC $\sim +23$ DR1 / $+28$ DR2). The comparison establishes kinematic shape-sufficiency, not statistical preference.

Model	k	joint χ^2	$\chi^2 - \chi^2_{\Lambda\mathrm{CDM}}$
free $E(z)$ spline (model-indep.)	spline	1391.85	-10.39
free-history TS ($f_v(z)$) *[this work]*	5	1396.06	-6.18
w_0w_aCDM	3	1398.29	-3.95
flat ΛCDM	1	1402.24	0
timescape tracker (1-param)	1	1469.29	$+67.06$

A no-lapse control (V0) gives $\chi^2 = 1396.49$, confirming the result is not an artifact of the lapse convention. An independent multistart re-derivation reproduces the headline $\chi^2 = 1396.06$ and cannot improve on it. Under the cosmology-independent Pantheon+ covariance of Ref. [18] (FLRW-fiducial bias correction removed) the shape-sufficiency fails: the fixed free-history exceeds best-fit ΛCDM by $\Delta(-2\ln\mathcal{L}) = +18/+24/+35$ at $z_{\min} = 0.055/0.033/0.010$ (all above the $+10$ threshold), $\Delta\mathrm{BIC}(\mathrm{FH} - \Lambda\mathrm{CDM}) = +11.1/+17.0/+27.2$ (fixed-history $k_{\mathrm{cosmo}} = 0$; $+45.8/+52.3/+64.0$ under an SN-fitted-nodes upper bound); the tracker reproduces Ref. [18]'s $\Delta\mathrm{BIC}(\mathrm{TS} - \Lambda\mathrm{CDM}) = +4.9/+1.5/-0.5$ (`seifert_covariance_row.json`).

1412.24 ($= \chi^2_{\Lambda\mathrm{CDM}}(\mathrm{DR1}) + 10$); *disfavoured* if ≤ 1427.24; *amplitude-dead* if ≥ 1469.29 (the tracker value). Part I is referenced to the DR1 ΛCDM baseline ($\chi^2_{\Lambda\mathrm{CDM}} = 1402.24$); Part II's model-comparison bars are referenced to the DR2 baseline ($\chi^2_{\Lambda\mathrm{CDM}} = 1399.81$).

The freed history reaches joint $\chi^2 = 1396.06$, comfortably inside the reconcile band and 6.18 below flat ΛCDM; only the model-independent free-$E(z)$ spline (1391.85) does better, and the rigid tracker (1469.29) is 67 worse. The preferred present void fraction is $f_{v0} = 0.640$ (LA), a physically plausible value, coincident with the BAO+CMB-preferred value of Paper I and with timescape's own Planck fits [13]. The associated Hubble rates are a dressed $H_0 = 63.97\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (g_{dress} convention), a full-rate dressed $67.65\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and a global bare $\bar{H}_0 = 53.79\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The pre-registered verdict is **RECONCILES (mechanism flexible)**; an independent re-derivation returns the same verdict. On DESI DR2 the free-history fit is $\chi^2 = 1398.28$ versus ΛCDM 1399.81 ($\Delta = -1.53$, Wilks $p = 0.82$), still inside the reconcile band; under BIC, ΛCDM is preferred in both DR1 and DR2—consistent with reading the reconciliation as kinematic sufficiency, not model preference.

The reconciliation is robust to the lapse convention.

TABLE II. Required void-fraction history $f_v^{\mathrm{req}}(z)$ at the fit nodes, for the three lapse readings, from the joint SN+BAO+CMB fit. The last column is the LA $\Delta\chi^2 \leq 1$ band. LA is the primary reading.

z	LA	LB	V0	LA $\Delta\chi^2 \leq 1$
0	0.640	0.588	0.383	$[0.638, 0.641]$
0.3	0.531	0.497	0.250	$[0.521, 0.533]$
0.7	0.396	0.385	0.128	$[0.384, 0.407]$
1.3	0.279	0.250	0.072	$[0.231, 0.318]$
2.33	0.194	0.121	0.017	$[0.148, 0.231]$

The LB (rate-ratio) sibling fits at $\chi^2 = 1406.92$ ($f_{v0} = 0.588$; $+4.69$ above ΛCDM, still inside the reconcile band), and its four validation gates all pass. The required-history nodes for all three lapse readings are collected in Table II: LA is the primary reading; LB and V0 bracket the systematic.

C. The amplitude split dissolves

Paper I's headline pathology was that the SN and BAO+CMB data, fitted through the rigid tracker, demand different present void-fraction *amplitudes*. We repeat that split test with the history freed, on DESI DR2 (a free-history joint refit fits at $\chi^2 = 1398.28$, $f_{v0} = 0.636$, versus ΛCDM 1399.81 and the tracker 1514.10). Fitting a single amplitude A to the SN sector and to the BAO+CMB sector separately, and jointly, gives $A_{\mathrm{SN}} = 0.6351$, $A_{\mathrm{BAO+CMB}} = 0.6363$, $A_{\mathrm{joint}} = 0.6360$, with a joint excess $\Delta\chi^2 = 0.027$—an amplitude split of $\boldsymbol{0.16\sigma}$. The same statistic applied to the tracker as a sanity check reproduces Paper I: $f_{v0}^{\mathrm{SN}} = 0.853$ against $f_{v0}^{\mathrm{BAO+CMB}} = 0.636$, $\Delta\chi^2 = 42.85$, a $\boldsymbol{6.5\sigma}$ split. The split *dissolves* once the history shape is free; an independent adversarial re-derivation confirms the dissolution at 0.16σ.

The interpretation is exactly Paper I's diagnosis, now demonstrated constructively: the SN and BAO+CMB data "want" the *same* amplitude once the history is free to bend. The 0.85-versus-0.64 split was the tracker parametrisation's rigidity, not a data conflict. One caveat bounds the claim: the free-history split constrains only the single overall amplitude A with the jointly-fit shape held fixed (the four extra shape degrees of freedom having already absorbed the SN-versus-BAO+CMB tension), so the 0.16σ demonstrates that the tracker's rigidity is removable, not that the data are intrinsically consistent under this model.

D. Why this is only a kinematic reconciliation

The reconciliation of Secs. III B–III C is a statement about distance geometry, not about dynamics. The forced $f_v(z)$ *violates the Buchert integrability condition.* For a genuine single-phase two-scale GR solution the void curvature parameter $\alpha^2 = -k_v f_{vi}^{2/3}$ must be constant in bare time (DNW13 Eq. 11 [12]); on the tracker it is constant to the $\sim 2\times10^{-6}$ pipeline floor, whereas the forced required history drifts by $\sim 152\%$ across its interior (Sec. IV C). The freed history is therefore "the backreaction the Hubble diagram wants"—a well-defined kinematic *target*, not a demonstrated general-relativistic solution. The geometric reconciliation is moreover specific to the supernova covariance: repeating the SN comparison under the cosmology-independent covariance of Refs. [17, 18]—which removes the FLRW-fiducial bias correction the timescape programme regards as circular—the fixed free-history is disfavoured relative to ΛCDM at every redshift cut (Table I, footnote), so the shape-sufficiency of Sec. III B is not robust to that choice. Part II asks, at full detail, whether that target is available in the sky, consistent as dynamics, and calibratable to the early Universe.

IV. PART II: DOES THE RECONCILIATION SURVIVE?

We report the four Part II tests at the same length and specificity as the Part I reconciliation. Each states its pre-registered threshold before its verdict. All four fail; the honest negative result is the value of the paper.

A. Availability: is the required history in the sky?

The reconciling history is a *prediction* about the void population: the required $f_v(z)$ must decline steeply with redshift. Over $0 < z < 2.33$ the LA required history declines by a total factor $\times 3.31$ (Table III). We confront this with the *observed* void volume fraction, measured from the BOSS DR12 galaxy sample with a watershed/ZOBOV void finder (Mao et al. 2017 [21]), calibrated to the $z = 0$ 2M++ field [26] and cross-checked against the watershed filling-fraction literature [22, 25]. The measurement chain is insensitive to the standard void observational systematics: the $z = 0$ anchor is the real-space-reconstructed 2M++ field; the $z > 0$ route uses the growth *ratio*, in which the redshift-space–distortion boost cancels to first order; the light-cone gradient across an individual void (its far wall seen $\sim 2\times10^8$ yr earlier) is $< 0.2\%$ in the volume fraction; and per-sightline propagation effects are bounded by Paper I's line-of-sight nulls ($\lesssim 0.05$ mag RMS).

The observed void fraction declines by a total factor of only $\times 1.16$ over the same range—the population is far flatter than the required history. At the pivotal node $z = 0.7$ the observed $f_v^{\rm obs} = 0.601$ against a required 0.396, a gap of 0.205 ($\sim 52\%$ relative). This gap is not a statistical undershoot but a *floor theorem*: the below-mean void fraction is bounded below by construction—the below-mean set is at least half the volume (≥ 0.5; empirically ≥ 0.56 across $R_s \in \{2, 4, 8\}\,h^{-1}$Mpc, giving the $z = 0.7$ band $[0.560, 0.654]$), whereas the required $f_v(0.7) = 0.396$ lies *below* that floor, so no admissible below-mean population can reach it. The gap is unbridgeable by construction, not a large-significance undershoot. The pre-registered SHAPE test declares the history *supplied* only if the observed decline ratio lies inside the required band at every node; here the observed ratio exceeds the required-hi edge at every node above zero. The $z = 0$ *level* matches (observed 0.643 inside $[0.614, 0.672]$; required $f_{v0} = 0.640$ passes), so it is specifically the *shape*, not the amplitude, that is unavailable.

Verdict: SHAPE-UNAVAILABLE. An independent re-derivation of the two-part telescope-forced test survives all four checks. A fit-space corroboration sharpens this: feeding the survey-measured history in with *zero* free parameters ($f_v(z) = f_v^{\rm obs}(z)$ fixed) gives a forced joint $\chi^2 = 2845.6$ on DR2, missing the (one-fewer-parameter) BIC bar [27] of 1407.2 by 1438; the miss is > 1400 under every covariance treatment tested, and an independent re-derivation passes 42/42 checks. No single fixed void definition simultaneously supplies the required $z = 0$ *level* and the required *decline*: the below-mean definition matches the level ($f_v(0) = 0.643$ in the band $[0.614, 0.672]$) but is only $\times 1.16$-flat, whereas a deep-void definition ($\delta < -0.3$) matches or exceeds the required $\times 3.31$ decline but starts too low ($f_v(0) \sim 0.44$–0.48, failing the level)—no single void definition occupies both corners. The $z = 0$ level-pass is moreover specific to the LA reading: LB ($f_{v0} = 0.588$) and V0 (0.383) fall below the below-mean band, so availability is worse under the lapse-reading systematic.

B. The local expansion-rate bias $b_{\rm pred}$

For the model to resolve the Hubble tension *as a local measurement bias*, its own two-scale structure must predict that a wall observer on the second rung of the SH0ES distance ladder measures an apparent H_0 high by a specific amount. Define the target from the model's internal ladder-versus-global bare Hubble offset,

$$b_{\rm req} = \frac{\bar{H}_0^{\rm anchored}}{\bar{H}_0^{\rm global}} - 1 = +8.4\% \quad (b_{\rm req} = 0.0842), \qquad (2)$$

which is the excess any local-bias mechanism must reproduce (the anchored fit that fixes the numerator passes an ΛCDM validation gate at $H_0 = 73.53 \pm 1.02\,{\rm km\,s^{-1}\,Mpc^{-1}}$, consistent with the SH0ES distance ladder [2]).

The apparent-H_0 window and the void-scale ceiling. The tracker-form combination

$$E_{\max}(f_{v0}) = \frac{3/2}{g_{\rm dress}(f_{v0})} - 1, \qquad (3)$$

with $g_{\rm dress}$ from Eq. (1), is an algebraic identity that, evaluated at Wiltshire's two published present void frac-

TABLE III. Required versus observed void-fraction decline. f_v^{req} is the LA required history (Table II); r_{req} is its pre-registered decline-ratio band $f_v(z)/f_v(0)$ (mapping forecast); f_v^{obs} and r_{obs} are the BOSS DR12 below-mean measurement at the required grid. The $z=0$ *level* matches (observed 0.643 inside the required band $[0.614, 0.672]$); it is the *shape* (decline) that is unavailable. The observed decline is flatter than the required band at every node above zero. Total decline $z=0 \to 2.33$: required $\times 3.31$, observed $\times 1.16$.

z	f_v^{req} (LA)	r_{req} band	f_v^{obs}	r_{obs}	status
0	0.640	— (anchor)	0.643	— (anchor)	LEVEL-PASS (band $[0.614, 0.672]$)
0.3	0.531	$[0.812, 0.836]$	0.623	0.968	FAIL: observed flatter
0.7	0.396	$[0.599, 0.637]$	0.601	0.934	FAIL: observed flatter (gap 0.205, 52%)
1.3	0.279	$[0.361, 0.499]$	0.578	0.898	FAIL (growth-extrap.): $r_{\mathrm{obs}} > r_{\mathrm{req}}^{\mathrm{hi}}$
2.33	0.194	$[0.231, 0.363]$	0.555	0.863	FAIL (growth-extrap.): $r_{\mathrm{obs}} > r_{\mathrm{req}}^{\mathrm{hi}}$

Observed f_v measured on the primary below-mean route at $z=0.3$ and $z=0.7$ (BOSS DR12, $R_s = 4\,h^{-1}$Mpc); the Rs-sensitivity band at $z=0.7$ is $[0.560, 0.654]$ across $R_s \in \{2,4,8\}\,h^{-1}$Mpc, all ≥ 0.5 (floor holds). At $z=1.3, 2.33$ the observed ratio exceeds the required-hi edge at every node (pre-registered SHAPE test).

tions, reproduces the 17–22% window as a consistency check (not an independent derivation): $E_{\max}(0.76) = 0.171$ and $E_{\max}(0.695) = 0.220$ (validation gate a, matching the literature window [7, 8, 15]). It is used only for this window check and does *not* give the ceiling adopted below—at the fitted $f_{v0} = 0.640$ it returns only 0.2615. The void-scale ceiling adopted below is a different quantity, the pure wall-observer clock conversion

$$E_{\max} = \bar{\gamma}_0\, H_{v0}/H_{\mathrm{dress},0} - 1 = 0.3365 \quad (\mathrm{LA};\ 0.208\ \mathrm{LB}), \tag{4}$$

evaluated on the fitted free history (gate b; this corrects an earlier mislabeling of the volume-average 0.195 as "the maximum"). The "maximum apparent-H_0 excess" is thus a modelling choice spanning a factor ~ 3 across conventions (tracker route 0.09–0.12; volume-average 0.195; void-scale maximum 0.337); we adopt the largest, model-favourable value, and the FAILS verdict below is robust to the choice.

The SH0ES geometry dilutes the ceiling. The second-rung Hubble-flow sample does not sit at the void-scale maximum. The survey-averaged bias is

$$b_{\mathrm{pred}}^{\mathrm{survey}} = E_{\max}\, \langle\varphi\rangle_{\mathrm{HF}} = 0.0240, \tag{5}$$

the void weight averaged over the $n_{\mathrm{HF}} = 277$ Hubble-flow SNe *alone* ($\langle\varphi\rangle_{\mathrm{HF}} = 0.0713$); the $n_{\mathrm{calib}} = 77$ calibrators ($\langle\varphi\rangle_{\mathrm{calib}} = 0.994$) do *not* enter, because the SH0ES M_B anchor is set by geometric Cepheid distances (parallax/maser/detached-eclipsing-binary), unbiased by the local apparent rate. This is a 14× dilution of the ceiling $E_{\max} = 0.3365$. Two bracketing *diagnostics* (neither is the primary): the differential $E_{\max}(\langle\varphi\rangle_{\mathrm{calib}} - \langle\varphi\rangle_{\mathrm{HF}}) = 0.310$ is physically inapplicable (the calibrators sit deep in the void), and the co-average over both populations, 0.092, has no physical basis. The lensing-*physical* void rate (calibrated to stacked void lensing densities [23, 24]) gives $b_{\mathrm{pred}}^{\mathrm{phys}} = 0.0158$, band $[0.0150, 0.0287]$. The entire admissible φ-band, $[0.0004, 0.0720]$, lies *below* $b_{\mathrm{req}} = 0.0842$: a one-sided under-prediction.

Verdict against the pre-registered threshold. The P3 threshold [1] is $|b_{\mathrm{pred}} - b_{\mathrm{req}}| \leq 1\sigma \Rightarrow$ RESOLVES; $\leq 2\sigma \Rightarrow$ PARTIAL; else FAILS. The decisive statement needs no σ: the entire admissible φ-band $[0.0004, 0.0720]$ lies below $b_{\mathrm{req}} = 0.0842$, so the mechanism under-predicts the required survey-averaged bias for *every* admissible φ (one-sided). Matching b_{req} would require a profile-decay scale $r_{\mathrm{hom}}^{*} = 140\,h^{-1}$Mpc, outside Wiltshire's declared $[80, 120]\,h^{-1}$Mpc band and beyond the $\sim 137.5\,h^{-1}$Mpc 2M++ homogeneity scale—where the density field is already homogeneous, so no apparent-H_0 bump can persist. Against the P3 tree the pre-registered significance is $n\sigma_{\mathrm{prereg}} = 1.57$ (PARTIAL), softened only by the wide φ-shape *theoretical* systematic, which the one-sided envelope shows *double-counts* σ_φ; the measurement-only $n\sigma_{\mathrm{meas}} = 4.16$ is the systematic-*excluded* bound, reported for completeness, not the basis of the verdict.

Catalog-forced cross-check (the symmetry rule). Substituting the *observed* void population (Sec. IV A) for the joint-fit winner makes the mechanism fail harder, not softer: the anchored ceiling drops to $E_{\max}^{\mathrm{obs}} = 0.0658$ and $b_{\mathrm{pred}}^{\mathrm{survey,obs}} = 0.0047$, while the forced global bare rate craters to $\bar{H}_0 = 50.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (missing the BIC bar by 1438). The under-prediction ratio $b_{\mathrm{req}}/b_{\mathrm{pred}}$ is 17.9× against the fitted target and 68.1× against the (larger) catalog-forced target. An independent re-derivation of both legs survives.

Verdict: PARTIAL (pre-registered) / FAILS (one-sided envelope + catalog-forced). The predicted local bias is an order of magnitude below the required +8.4% whether the void structure is fitted to the Hubble diagram or supplied by the telescope. The family-level kill rests on the one-sided envelope above and the catalog-forced FAILS, reached through the pre-registered decision tree—not on relabeling the 4.16σ measurement-only significance as "robust". It kills the local-bias resolution for this model family regardless of the joint-fit quality of Part I.

C. Dynamical consistency: is the required history a GR solution?

Section III D noted that the forced history violates Buchert integrability; here we quantify it and make the honest attempt to *restore* consistency. The invariant $\alpha^2 = -k_v\, f_{vi}^{2/3}$ (DNW13 Eq. 11 [12]) must be constant in bare time for a genuine single-phase two-scale GR solution. Its fractional drift is $\sim 152\%$ across the interior of the forced required history and $\sim 5\times$ (span) for the observed history, against the tracker's 2×10^{-6} pipeline floor—six-plus orders of magnitude worse. An exact-analytic transcription check drifts only at machine precision, so the drift is physical, not numerical. *No consistent single-phase Buchert model tracks either history.*

The honest attempt to make it consistent is a *three-phase* Buchert solve (shallow void, deep void, wall), which adds the dynamical degrees of freedom that a single phase lacks. All four solver gates pass—G1 tracker-limit (SN $\chi^2 = 1391.545$), G2 integrability, G3 lapse-limit (agreement with $\frac{1}{2}(2+f_v)$ to 4×10^{-16}), G4 numerics—and the stage verdict is GO. But the best fit is *degenerate*: the extra void phase's amplitude drives to $\alpha_{s2} \to 0$, i.e. the three-phase model collapses to the two-phase empty-void tracker with a single genuine void of fraction $f_{d0} \simeq 0.27$. The tracking solution "tracks" only by becoming the tracker.

The pre-registered geometry bar (DR2) is $\chi^2 \leq \chi^2_{\Lambda\mathrm{CDM}}(\mathrm{DR2}) + \ln N \approx 1407.2$. The zero-parameter ($k=0$) forced-geometry prediction of the best three-phase constants gives a joint $\chi^2 = 6360.2$, missing the bar by **4953**; a degeneracy check confirms this equals the tracker prediction at $f_{v0} = f_{d0} = 0.274$ to 0.02 in χ^2. Even the never-a-claim *constants-freed* diagnostic ceiling—which frees parameters the theory does not permit—misses the bar by 107 ($\Delta\mathrm{BIC} = +136$ versus ΛCDM).

Verdict: CLOSED (a theory-level close-out, not a numerical failure). The required history is not a general-relativistic single-phase solution, and the dynamical three-phase attempt to build one collapses back to the tracker it was meant to generalise.

D. The sound horizon: does it survive to the early Universe?

Part I's fit borrowed the external drag-epoch sound horizon $r_d = 147.09$ Mpc as a ruler. WP-C replaces that crutch with the model's own bare-frame radiation-era physics (DNW13 machinery [12]) and asks what r_d free-history timescape actually predicts. The answer is

$$r_d^{\text{in-model}} = 199.6 \text{ Mpc}, \tag{6}$$

$+35.7\%$ above the external drag-epoch ruler $r_d = 147.09$ Mpc (the in-model decoupling-epoch radius is $r_{\mathrm{dec}} = 195.9$ Mpc). The machinery is validated: its ΛCDM gate returns $r_\star = 144.3$ Mpc and $r_{\mathrm{drag}} = 146.9$ Mpc, matching the expected values.

Part I's distance fit (Sec. III B) is untouched by the in-model r_d: its r_d-independence comes from the analytic α-marginalization of the BAO+CMB amplitude (Sec. III A), which absorbs any overall ruler rescaling—*not* from the near-unity acoustic ratio. The in-model geometry nonetheless disagrees with CMB early-Universe physics: although the acoustic ratio $r_\star/r_{\mathrm{drag}}$ matches Planck to 0.03% (0.9816 vs 0.9819), the in-model acoustic angle $\ell_A = 218.9$ fails Planck's 301.7 by 27% (291.9 vs 301.7 even when rescaled by the external r_d), with bare shift $R = 1.11$ vs 1.74 and a BBN-η baryon density $\omega_b = 0.0186$ vs 0.0224. The Part I CMB point therefore constrains only the redshift *shape*, not agreement with CMB early-Universe physics. But the *absolute* Hubble rate is not r_d-independent. With the in-model r_d, the global bare rate drops from $\bar{H}_0 = 53.8$ to $39.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the self-consistent fixed point is $\bar{H}_0^* = 32.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (at $r_d^* = 244.5$ Mpc), with the corresponding dressed rates falling to 47.2 (g_{dress}) and $49.9\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (full-rate)—far below both the reconciling band and SH0ES.

The pre-registered question is whether the Part I *absolute-H_0* claim survives an in-model sound horizon; it does not ($R1_{\mathrm{survives}} = \mathrm{false}$). An independent re-derivation reproduces every headline number. As in Paper I, we distinguish resolution from recasting: driving the bare Hubble rate *below* both canonical values is a recast of the tension, not a reconciliation near the local H_0.

Verdict: FAILS (as an absolute-H_0 claim). The model's own early-Universe physics predicts a sound horizon $\sim 36\%$ larger than the ruler it had borrowed; forcing self-consistency collapses the bare Hubble rate to ~ 32–$40\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

E. Self-inflicted tensions (WP-N)

Table IV aggregates the program's tension ledger: for each tension, the free-history (kinematic) status, whether it is *self-inflicted* by the kinematic reconciliation, and the source artifact. The tally is 6 YES self-inflicted, 2 partial, and 1 target (the Hubble tension itself, which the model exists to resolve). The synthesis is that the mechanism is CLOSED at every level tested: the void geometry is shape-unavailable, the local SH0ES bias fails, the dynamical three-phase consistency is closed, and the sound-horizon absolute-H_0 reconciliation fails.

The growth/S_8 row deserves separate note: it is *undefined-in-kinematic-reading*. The kinematic reconciliation has no perturbed metric, so there is no growth equation on the averaged background—$f\sigma_8$, RSD β, weak-lensing/S_8, cluster counts, and the ISW are none of them rigorously defined. The measured $\sigma_m(L = 20\,h^{-1}\mathrm{Mpc}, z)$ count-in-cells amplitude ($A = 0.444$, reduced $\chi^2 \approx 0.22$) is an *input*, not a prediction. This is a structural boundary of the kinematic reading, not a

TABLE IV. WP-N self-inflicted tensions of the free-history (kinematic) reading. "Self-inflicted?" marks tensions created *by* the kinematic reconciliation rather than inherited. Tally: 6 YES, 2 partial, 1 target. The growth/S_8 row (8) is structurally, not numerically, undefined: with no perturbed metric the kinematic reading cannot even pose the S_8 question.

Tension	Free-history (kinematic) status	Self-infl.?	Source
Hubble (SH0ES vs. early)	FAILS: $b_{\rm pred}^{\rm survey}$ ceiling 0.0240 + physical 0.0158 vs. $b_{\rm req} = 0.0842$ (meas.-only $n\sigma = 4.16$ ceiling / 4.73 physical)	no (target)	`bpred_survey_averaged.json`
Required-vs-observed void shape (floor)	SHAPE-UNAVAILABLE (obs decline $\times 1.16$ vs. required $\times 3.31$; floor theorem)	YES	`R2_final.json`
WP-B dynamical three-phase	CLOSED: TRACKS degenerately but $k = 0$ geometry misses BIC bar by 4953; T1 ceiling misses by 107 (ΔBIC+136)	YES	`threephase_forced_geometry.json`
Buchert integrability	Not a single-phase GR solution: α^2 drifts $\sim 152\%$ (forced, interior) / $\sim 5\times$ (observed, span) vs. tracker $\sim 2 \times 10^{-6}$	YES	`wpb_integrability.json`
Void–wall contrast budget (two-sided)	MARGINAL (field-mean 0.375–0.401 $<$ required 0.474 at $z = 0$; deep-void+wall 0.46–0.57 reaches/exceeds it)	partial	`contrast_budget.json`
Q-budget (Stage 0)	GATE=GO on empty/kinematic ceiling; physical budget 4–8$\times$ short; required contrast exceeds empty-Milne ceiling $\times 1.36$	YES	`q_budget.json`
Sound horizon / r_d	FAILS: in-model $r_d = 199.6$ Mpc (+36%); implied bare $\bar{H}_0$ 53.8 $\to\sim$40 (fixed point 32.4)	YES	`wpc_sound_horizon.json`
Growth / S_8	UNDEFINED-IN-KINEMATIC-READING (no perturbed metric $\Rightarrow$ no growth equation; measured $\sigma_m(z)$ is an input, not a prediction)	YES	`NOTES_growth.md`
$b_{\rm pred}$ z-profile (P4)	predicts local H_0 bump decaying by $r_{\rm hom} \sim 100\,h^{-1}$Mpc; $r^*_{\rm hom} = 140$ to reach $b_{\rm req}$ is outside declared $[80, 120]$ and beyond 2M++ homogeneity $\sim$137.5	partial	`bpred_survey_averaged.json`

computed defeat, and it is defined only after the (closed) dynamical WP-B reading is built.

V. DISCUSSION

The result is *necessary but not sufficient*, made concrete. Paper I's diagnosis was correct: the SN-versus-BAO+CMB split is the tracker's one-parameter rigidity, and removing that rigidity by freeing $f_v(z)$ reconciles the distance geometry below ΛCDM at a plausible $f_{v0} = 0.640$, with the amplitude split collapsing from 6.5σ to 0.16σ (Sec. III). Yet the reconciling history is kinematic, and the kinematic/dynamic gap is the whole story: the history is *not available* in the observed void population (Sec. IV A; the required $\times 3.31$ decline against an observed $\times 1.16$, unbridgeable by a floor theorem), *not a GR solution* (Sec. IV C; the three-phase Buchert attempt collapses to the tracker and misses the geometry bar by 4953), *not early-Universe calibratable* (Sec. IV D; an in-model r_d that is +36% too large drives the bare rate to $\sim$32–40), and predicts *the wrong local bias* (Sec. IV B; +2.4% against the required +8.4%, with the full admissible band below the target—pre-registered PARTIAL, but failing outright via the one-sided envelope and the catalog-forced test).

A *positive* result would have required, jointly and specifically: an observed void population steep enough to supply the $\times 3.31$ decline (not the observed $\times 1.16$); a required history that is a single-phase integrability-consistent Buchert solution (not one drifting $\sim 152\%$ in α^2); and an in-model sound horizon near 147 Mpc (not 199.6). None of these holds, and the local-bias prediction independently falls an order of magnitude short. The distinction between resolution and recasting from Paper I recurs sharply here: even where the model can be pushed toward a lower absolute Hubble rate, that is a recast of the tension—both canonical values reinterpreted as biased relative to a lower global average—not a reconciliation near the local H_0.

The S_8/growth question is out of reach of the kinematic reading altogether: it has no perturbed metric and hence no growth equation, so it cannot pose the question, let alone answer it (Sec. IV E). We flag this as the boundary of the kinematic reading, not as a computed defeat; a dynamical reading that could pose it is exactly the (closed) three-phase construction of Sec. IV C.

VI. CONCLUSIONS

Free-history timescape does not resolve the Hubble tension. Freeing the void-fraction history from the one-parameter tracker removes the rigidity that Paper I identified and reconciles the SN+BAO+CMB distance geometry at joint $\chi^2 = 1396.1$, below flat ΛCDM (1402.2) and far below the tracker (1469.3), at a physically plausible $f_{v0} = 0.640$, with the amplitude split dissolving to 0.16σ (and even this geometric reconciliation is covariance-dependent: under the cosmology-independent supernova covariance of Refs. [17, 18] the fixed free-history is disfavoured relative to ΛCDM at every redshift cut). But that flexibility is kinematic, and it closes at every deeper physical test: availability (shape-unavailable, decline ×3.31 required vs ×1.16 observed), local bias (+2.4% predicted vs +8.4% required—the full admissible band below the target; pre-registered PARTIAL, failing outright when the observed catalog is forced in), dynamical consistency (closed; the three-phase Buchert solve collapses to the tracker and misses the geometry bar by 4953), and the sound horizon (fails; in-model $r_d = 199.6$ Mpc, +36%, driving the bare rate to ~ 32–$40\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$). Reconciliation without resolution. This corroborates and sharpens Paper I: on present public data, eliminating dark energy through inhomogeneity—whether through the rigid tracker or its fully freed history—does not resolve the Hubble tension.

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DATA AND CODE AVAILABILITY

Every quantitative claim reproduces from a committed script under the discipline *one number* → *one script* → *one result file*, with pre-registered thresholds fixed before each verdict. The supernova data are the public Pantheon+ release [19, 20]; the BAO and CMB inputs are DESI DR2 [5] and the Planck 2018 acoustic scale [3]; the observed void population is from BOSS DR12 [21] and the 2M++ field [26]. This repository, `free-history-timescape`, holds the Part I geometry-reconciliation code at the root and the Part II tension-test code under `tensions/`; the former `free-history-timescape-tensions` repository is archived read-only on GitHub as the per-commit history of record. The code is released under the MIT license and will be deposited publicly on publication.

[1] V. Zhygulin, "Can eliminating dark energy resolve the Hubble tension? A uniform test of void and backreaction cosmologies against supernovae, baryon acoustic oscillations, and the CMB acoustic scale" (2026), companion paper (Paper I of this program).

[2] A. G. Riess *et al.*, Astrophys. J. Lett. **934**, L7 (2022), arXiv:2112.04510.

[3] Planck Collaboration, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[4] DESI Collaboration (2024), arXiv:2404.03002.

[5] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints" (2025), arXiv:2503.14738.

[6] D. L. Wiltshire, Phys. Rev. Lett. **99**, 251101 (2007), arXiv:0709.0732.

[7] D. L. Wiltshire, Phys. Rev. D **80**, 123512 (2009), arXiv:0909.0749.

[8] D. L. Wiltshire (2009), arXiv:0912.5234.

[9] T. Buchert, Gen. Relativ. Gravit. **32**, 105 (2000), arXiv:gr-qc/9906015.

[10] A. Wiegand, T. Buchert (2010), arXiv:1002.3912.

[11] B. M. Leith, S. C. C. Ng, D. L. Wiltshire, Astrophys. J. Lett. **672**, L91 (2008), arXiv:0709.2535.

[12] J. A. G. Duley, M. A. Nazer, D. L. Wiltshire, Class. Quantum Grav. **30**, 175006 (2013), arXiv:1306.3208.

[13] M. A. Nazer, D. L. Wiltshire, Phys. Rev. D **91**, 063519 (2015), arXiv:1410.3470.

[14] P. R. Smale, D. L. Wiltshire, Mon. Not. R. Astron. Soc. **413**, 367 (2011), arXiv:1009.5855.

[15] D. L. Wiltshire, P. R. Smale, T. Mattsson, R. Watkins, Phys. Rev. D **88**, 083529 (2013), arXiv:1201.5371.

[16] L. H. Dam, A. Heinesen, D. L. Wiltshire, Mon. Not. R. Astron. Soc. **472**, 835 (2017), arXiv:1706.07236.

[17] Z. G. Lane, A. Seifert, R. Ridden-Harper, D. L. Wiltshire, Mon. Not. R. Astron. Soc. **536**, 1752 (2025), arXiv:2311.01438.

[18] A. Seifert, Z. G. Lane, M. Galoppo, R. Ridden-Harper, D. L. Wiltshire, Mon. Not. R. Astron. Soc. Lett. **537**, L55 (2025), arXiv:2412.15143.

[19] D. Scolnic *et al.*, Astrophys. J. **938**, 113 (2022), arXiv:2112.03863.

[20] D. Brout *et al.*, Astrophys. J. **938**, 110 (2022), arXiv:2202.04077.

[21] Q. Mao *et al.*, Astrophys. J. **835**, 161 (2017), arXiv:1602.02771.
[22] D. C. Pan, M. S. Vogeley, F. Hoyle, Y.-Y. Choi, C. Park, Mon. Not. R. Astron. Soc. **421**, 926 (2012), arXiv:1103.4156.
[23] R. K. Sheth, R. van de Weygaert (2004), arXiv:astro-ph/0311260.
[24] J. Clampitt, B. Jain (2015), arXiv:1404.1834.
[25] M. J. Williams, H. J. Macpherson, D. L. Wiltshire, C. Stevens, Mon. Not. R. Astron. Soc. **536**, 2645 (2025), arXiv:2403.15134.
[26] J. Carrick, S. J. Turnbull, G. Lavaux, M. J. Hudson, Mon. Not. R. Astron. Soc. **450**, 317 (2015), arXiv:1504.04627.
[27] R. E. Kass, A. E. Raftery, J. Am. Stat. Assoc. **90**, 773 (1995).